

About OpenAI

- What is OpenAI
- OpenAI Models
- What is Token?

What is Open AI

- OpenAI, founded in **December 2015** is a private research organization and technology company that develops **AI-powered products** and tools that benefits humanity:
- It is committed to ensure AI is developed and used **Ethical** to mitigate potential risks.
- Originally a non-profit, OpenAI transitioned to a “**capped-profit**” model to attract investment for large-scale AI projects. Balances research with developing market-leading AI solutions like GPT and DALL·E.

Founders: Elon Musk, Sam Altman, Greg Brockman.

OpenAI has partnered with Microsoft is a Strategic partnership with a \$1 billion investment. Integration of OpenAI models into Azure AI services.

Types of OpenAI Models

1. Frontier / Flagship Models

- Most advanced reasoning, understanding, instruction-following, multimodal support, large context windows, best for complex tasks, coding, agentic workflows.
- Strong general intelligence and versatile across domains (text, code, vision, etc.).

Example Models:

- GPT-5.2 (best overall; excels at reasoning & coding)
- GPT-5.2-pro (higher precision & more compute)
- GPT-5.1 family (previous flagship)
- GPT-5 (earlier flagship)

2. General-Purpose / Instruction Models

- Great for conversational AI, instruction following, summarization, Q&A, and broad usage.
- Large context windows for longer inputs.

Example Models:

- GPT-4.1 (high-quality text & instruction tasks)
- GPT-4.1 mini & nano (more cost-efficient / faster variants)

3. Reasoning / Task-Focused Models

- Enhanced reasoning depth and problem solving — useful for science, math, logic, and complex decisions.
- Often optimized for multi-step chain of thought reasoning.

Example Models:

- o3 (deep reasoning model)
- o4-mini (fast & affordable reasoning)
- o3-mini (cost-efficient reasoning)

4. Realtime & Audio/Multimodal Models

- Support real-time text & audio processing, speech-to-text, text-to-speech, and multimodal interactions.

Example Models:

- gpt-realtime (real-time text & audio)
- gpt-realtime-mini (cost-efficient realtime)
- gpt-audio & gpt-audio-mini (audio-centric)
- GPT-4o Audio & mini variants (voice + text)

5. Vision & Image Models

- Vision understanding, image analysis, and generation. Useful for multimodal AI apps involving pictures.

Example Models:

- DALL-E 3 (current image-generation model)
- GPT Image 1.5 (updated image generation)
- chatgpt-image-latest (image model in ChatGPT)

6. Embeddings Models

- Produce vector embeddings for search, clustering, similarity, retrieval, classification.
- Used widely in semantic search and recommendation systems.

Example Models:

- text-embedding-3-large (most capable)
- text-embedding-3-small (cost-efficient)

7. Open-Weight / Research Models

- Models with weights released openly to enable deployment on personal hardware and research fine-tuning.
- Designed for broader accessibility and experimentation.

Example Models:

- gpt-oss-120b (high-performance open model)
- gpt-oss-20b (smaller, mobile/edge use)

8. Specialized Tool & Moderation Models

- Specific roles like content moderation, real-time computer interface tasks, specialized API workflows.

Example Models:

- omni-moderation (content moderation)
- computer-use-preview (tools & workflows)

9. Open Weight Models by OpenAI:

- Designed for agentic tasks: Leverage powerful instruction following and tool use within the chain-of-thought, including web search and Python code execution.
- Deeply customizable: Adjust the reasoning effort to low, medium, or high. Plus, customize the models to adapt to your use case with full-parameter fine-tuning.
- Playground: <https://gpt-oss.com/>
- Full chain-of-thought: Access the full chain-of-thought for easier debugging and higher trust in model outputs.

Example Models:

- gpt-oss-120b
- gpt-oss-20b

Choosing a model

A key choice to make when generating content through the API is which model you want to use –

- **Reasoning models** generate an internal chain of thought to analyze the input prompt, and excel at understanding complex tasks and multi-step planning. They are also generally slower and more expensive to use than GPT models.
- **GPT models** are fast, cost-efficient, and highly intelligent, but benefit from more explicit instructions around how to accomplish tasks.

- **Large and small (mini or nano) models** offer trade-offs for speed, cost, and intelligence. Large models are more effective at understanding prompts and solving problems across domains, while small models are generally faster and cheaper to use.

Models Documentation:

<https://platform.openai.com/docs/models/>

Comparing Models:

<https://platform.openai.com/docs/models/compare>

What is Token?

Token is a **unit of text** that the model processes. It can be as small as a single character or as large as a whole word, depending on the **context** and **language**.

Tokens are not cut up exactly where the words start or end - tokens can include trailing spaces and even sub-words.

The sentence “**ChatGPT is great!**” is broken into tokens like:

- “Chat”, “GPT”, “ is”, “ great”, “!”.

Here are some helpful rules of thumb for understanding tokens in terms of lengths:

- 1 token $\approx$ 4 chars in English $\approx$ $\frac{3}{4}$ words
- 100 tokens $\approx$ 75 words

Or

- 1-2 sentence $\approx$ 30 tokens
- 1 paragraph $\approx$ 100 tokens
- 1,500 words $\approx$ 2048 tokens

How words are split into tokens is also language-dependent. For example ‘Cómo estás’ (*‘How are you’* in Spanish) contains 5 tokens (for 10 chars) आप कैसे हैं? Contains 3 tokens (11 chars). The higher token-to-char ratio can make it more expensive to implement the API for languages other than English.

Tokenizer tool (<https://platform.openai.com/tokenizer>), allows you to calculate the number of tokens

Please note that the exact tokenization process varies between models.

Alternatively, if you'd like to tokenize **text programmatically**, use [Tiktoken](#) as a fast BPE tokenizer specifically used for OpenAI models.

Token Types:

- **Prompt tokens** are the tokens that you input into the model. This is the number of tokens in your prompt.
- **Completion tokens** are any tokens that the model generates in response to your input. For a standard request, this is the number of tokens in the completion.

Pricing Calculation based on tokens:

<https://platform.openai.com/docs/pricing>

Rate Limit:

<https://platform.openai.com/docs/guides/rate-limits>